DAILY PRACTICE CALENDAR

AP Calculus AB • 2027–2028

34 weeks • Monday–Friday • 10 problems a day (about 30–40 minutes)

Experience first. Symbols second. Understanding always.

How to Use This Calendar

- **Mon–Thu:** 10 problems on the day's skill: **8 new + 2 review** from earlier weeks. Start with the **Try** problem as the warm-up.
- **Friday:** Weekly Quiz of 10 mixed problems (8 this week + 2 review). Copy every missed problem into the **error log** and redo it next Monday.
- **Holidays and breaks** are marked. A holiday's skill moves to the weekend; break weeks are not counted in the 34 weeks.
- This calendar follows the **AP Calculus AB** course framework and covers AP Units 1–8: limits, derivatives, applications of derivatives, integrals, differential equations and applications of integration. Answers to every **Try** problem are on the last page.

Year at a Glance

Wk	Dates	Topic
1	Aug 16–Aug 20	Limits from Graphs & Tables
2	Aug 23–Aug 27	Limit Laws & Algebra
3	Aug 30–Sep 3	Continuity & the Intermediate Value Theorem
4	Sep 6–Sep 10	Limits at Infinity & the Squeeze Theorem
5	Sep 13–Sep 17	Defining the Derivative
6	Sep 20–Sep 24	Power, Product & Quotient Rules
7	Sep 27–Oct 1	Trig, Exponential & Log Derivatives
8	Oct 4–Oct 8	The Chain Rule
9	Oct 11–Oct 15	Implicit & Inverse Derivatives
10	Oct 18–Oct 22	Review & Checkpoint 1
11	Oct 25–Oct 29	Rates of Change in Context & Motion
12	Nov 1–Nov 5	Related Rates
13	Nov 8–Nov 12	Linearization & L'Hôpital's Rule
14	Nov 15–Nov 19	The Mean Value Theorem & Extrema
	Nov 22–Nov 26	*Thanksgiving Break*
15	Nov 29–Dec 3	Concavity & the Second Derivative Test
16	Dec 6–Dec 10	Optimization & Curve Sketching
17	Dec 13–Dec 17	Riemann Sums & Accumulation
	Dec 20–Dec 24	*Winter Break*
	Dec 27–Dec 31	*Winter Break*
18	Jan 3–Jan 7	The Fundamental Theorem of Calculus
19	Jan 10–Jan 14	Antiderivatives & Indefinite Integrals
20	Jan 17–Jan 21	Mid-Year Review & Checkpoint
21	Jan 24–Jan 28	Integration by Substitution
22	Jan 31–Feb 4	Accumulation Functions & Integral Properties
23	Feb 7–Feb 11	Slope Fields & Separable Equations
24	Feb 14–Feb 18	Exponential Models
25	Feb 21–Feb 25	Average Value & Motion with Integrals
26	Feb 28–Mar 3	Area Between Curves
27	Mar 6–Mar 10	Volumes with Cross Sections
28	Mar 13–Mar 17	Volumes: Disc & Washer Methods
29	Mar 20–Mar 24	Accumulation in Context
	Mar 27–Mar 31	*Spring Break*
30	Apr 3–Apr 7	Free-Response Practice: Tables & Graphs
31	Apr 10–Apr 14	Multiple-Choice Practice
32	Apr 17–Apr 21	AP Review: Limits & Derivatives
33	Apr 24–Apr 28	AP Review & Practice Exam
34	May 1–May 5	Final Checkpoint & Celebration

Units 1 & 6 · Unit 2 · Unit 3 · Unit 4 · Unit 5 · Wrap-Up · Break

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WEEK 1 • Aug 16–Aug 20, 2027 — Unit 1: Limits & Continuity

Limits from Graphs & Tables

Core: A limit is where the outputs are heading, not necessarily where they are.

Mon Aug 16
Limits from tables
TRY x: 1.9, 1.99, 2.01, 2.1 and $f(x)$: 3.8, 3.98, 4.02, 4.2. Estimate $\lim_{x\to 2} f(x)$.

Tue Aug 17
One-sided limits
TRY For $f(x) = \frac{|x|}{x}$, find $\lim_{x\to 0^-} f(x)$ and $\lim_{x\to 0^+} f(x)$.

Wed Aug 18
When a limit fails
TRY Does $\lim_{x\to 0} \frac{|x|}{x}$ exist?

Thu Aug 19
Infinite limits
TRY Find $\lim_{x\to 3^+} \frac{1}{x-3}$.

Fri Aug 20
Weekly Quiz
10 mixed problems: 8 on this week's skills + 2 review. Correct and log every miss.

WEEK 2 • Aug 23–Aug 27, 2027 — Unit 1: Limits & Continuity

Limit Laws & Algebra

Core: Try substitution first; if you get $\frac{0}{0}$, factor, rationalize or use a known limit.

Mon Aug 23
Direct substitution
TRY Find $\lim_{x\to 2}(x^2 + 3x)$.

Tue Aug 24
Factoring
TRY Find $\lim_{x\to 3} \frac{x^2-9}{x-3}$.

Wed Aug 25
Rationalizing
TRY Find $\lim_{x\to 0} \frac{\sqrt{x+4}-2}{x}$.

Thu Aug 26
Special trig limits
TRY Find $\lim_{x\to 0} \frac{\sin 3x}{x}$.

Fri Aug 27
Weekly Quiz
10 mixed problems: 8 on this week's skills + 2 review. Correct and log every miss.

WEEK 3 • Aug 30–Sep 3, 2027 — Unit 1: Limits & Continuity

Continuity & the Intermediate Value Theorem

Core: f is continuous at a when $\lim_{x\to a} f(x) = f(a)$.

Mon Aug 30
Checking continuity
TRY $f(x) = x^2$ for $x < 1$ and $f(x) = 2x - 1$ for $x \geq 1$. Is f continuous at $x = 1$?

Tue Aug 31
Making it continuous
TRY $f(x) = kx + 1$ for $x < 2$ and $f(x) = x^2$ for $x \geq 2$. Find k so f is continuous.

Wed Sep 1
Types of discontinuity
TRY What kind of discontinuity does $\frac{x^2-1}{x-1}$ have at $x = 1$?

Thu Sep 2
The IVT
TRY f is continuous, $f(1) = -2$ and $f(3) = 5$. Must $f(c) = 0$ for some c in $(1, 3)$?

Fri Sep 3
Weekly Quiz
10 mixed problems: 8 on this week's skills + 2 review. Correct and log every miss.

WEEK 4 • Sep 6–Sep 10, 2027 — Unit 1: Limits & Continuity

Limits at Infinity & the Squeeze Theorem

Core: Compare the highest powers to find the end behavior.

Mon Sep 6
HOLIDAY: Labor Day
No class. Do this skill's 10 problems on the weekend:
Limits at infinity
TRY Find $\lim_{x\to\infty} \frac{3x^2-x}{x^2+4}$.

Tue Sep 7
Comparing degrees
TRY Find $\lim_{x\to\infty} \frac{x+1}{x^2+1}$.

Wed Sep 8
Squeeze theorem
TRY Find $\lim_{x\to 0} x^2 \sin\frac{1}{x}$.

Thu Sep 9
Exponential end behavior
TRY Find $\lim_{x\to-\infty} e^x$.

Fri Sep 10
Weekly Quiz
10 mixed problems: 8 on this week's skills + 2 review. Correct and log every miss.

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WEEK 5 • Sep 13–Sep 17, 2027 — Unit 2: Differentiation

Defining the Derivative

Core: $f'(x) = \lim_{h\to 0} \frac{f(x+h)-f(x)}{h}$: the instantaneous rate of change.

Mon Sep 13
Average vs. instantaneous
TRY For $f(x) = x^2$, find the average rate of change on $[1, 1.1]$.

Tue Sep 14
The limit definition
TRY Use the limit definition to find $f'(x)$ for $f(x) = x^2$.

Wed Sep 15
Tangent lines
TRY Find the tangent line to $y = x^2$ at $x = 3$.

Thu Sep 16
Differentiability
TRY Is $f(x) = |x|$ differentiable at $x = 0$?

Fri Sep 17
Weekly Quiz
10 mixed problems: 8 on this week's skills + 2 review. Correct and log every miss.

WEEK 6 • Sep 20–Sep 24, 2027 — Unit 2: Differentiation

Power, Product & Quotient Rules

Core: $(f \cdot g)' = f'g + fg'$ and $\left(\frac{f}{g}\right)' = \frac{f'g-fg'}{g^2}$.

Mon Sep 20
Power rule
TRY Find $\frac{d}{dx}\left(4x^3 - 2\sqrt{x}\right)$.

Tue Sep 21
Product rule
TRY Use the product rule to find $\frac{d}{dx}\left[(x^2+1)(x^3-x)\right]$.

Wed Sep 22
Quotient rule
TRY Find $\frac{d}{dx}\frac{x}{x+1}$.

Thu Sep 23
Using a table
TRY $f(2) = 3$, $f'(2) = -1$, $g(2) = 4$ and $g'(2) = 5$. Find $(f \cdot g)'(2)$.

Fri Sep 24
Weekly Quiz
10 mixed problems: 8 on this week's skills + 2 review. Correct and log every miss.

WEEK 7 • Sep 27–Oct 1, 2027 — Unit 2: Differentiation

Trig, Exponential & Log Derivatives

Core: $(\sin x)' = \cos x$, $(e^x)' = e^x$, $(\ln x)' = \frac{1}{x}$.

Mon Sep 27
Sine & cosine
TRY Find $\frac{d}{dx}(\sin x + \cos x)$.

Tue Sep 28
Tangent
TRY Find $\frac{d}{dx}\tan x$.

Wed Sep 29
Exponentials
TRY Find $\frac{d}{dx}(x \cdot e^x)$.

Thu Sep 30
Logarithms
TRY Find the derivative of $\ln x$ at $x = 4$.

Fri Oct 1
Weekly Quiz
10 mixed problems: 8 on this week's skills + 2 review. Correct and log every miss.

WEEK 8 • Oct 4–Oct 8, 2027 — Unit 2: Differentiation

The Chain Rule

Core: $\frac{d}{dx}f(g(x)) = f'(g(x)) \cdot g'(x)$: outside, then inside.

Mon Oct 4
Powers
TRY Find $\frac{d}{dx}(3x+1)^5$.

Tue Oct 5
Trig
TRY Find $\frac{d}{dx}\sin(x^2)$.

Wed Oct 6
Exponentials
TRY Find $\frac{d}{dx}e^{-2x}$.

Thu Oct 7
Using a table
TRY $f(1) = 2$, $f'(1) = 3$ and $g'(2) = 4$. If $h(x) = g(f(x))$, find $h'(1)$.

Fri Oct 8
Weekly Quiz
10 mixed problems: 8 on this week's skills + 2 review. Correct and log every miss.

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WEEK 9 • Oct 11–Oct 15, 2027 — Unit 2: Differentiation

Implicit & Inverse Derivatives

Core: Differentiate both sides with respect to x; every y gets a $\frac{dy}{dx}$.

Mon Oct 11
Implicit differentiation
TRY Find $\frac{dy}{dx}$ for $x^2 + y^2 = 25$.

Tue Oct 12
Tangent slopes
TRY Find the slope of $x^2 + y^2 = 25$ at $(3, 4)$.

Wed Oct 13
Inverse trig
TRY Find $\frac{d}{dx} \arctan x$.

Thu Oct 14
Inverse functions
TRY $f(2) = 5$ and $f'(2) = 4$. Find $(f^{-1})'(5)$.

Fri Oct 15
Weekly Quiz
10 mixed problems: 8 on this week's skills + 2 review. Correct and log every miss.

WEEK 10 • Oct 18–Oct 22, 2027 — Unit 2: Differentiation

Review & Checkpoint 1

Core: Pull Units 1–2 together and fix any weak spots.

Mon Oct 18
Review: limits
TRY Find $\lim_{x \to 0} \frac{1 - \cos x}{x}$.

Tue Oct 19
Review: product rule
TRY Find $\frac{d}{dx}\left(x^3 \sin x\right)$.

Wed Oct 20
Review: chain rule
TRY Find $\frac{d}{dx} \ln(x^2 + 1)$.

Thu Oct 21
Review: tangent lines
TRY Find the tangent line to $y = e^x$ at $x = 0$.

Fri Oct 22
Checkpoint 1
Units 1–2 test. Afterward, start an error log of every missed problem.

WEEK 11 • Oct 25–Oct 29, 2027 — Unit 3: Applying Derivatives

Rates of Change in Context & Motion

Core: A derivative is a rate: always give its meaning with units.

Mon Oct 25
Meaning & units
TRY $V(t)$ is liters of water at t minutes, and $V'(5) = -3$. What does this mean?

Tue Oct 26
Velocity
TRY Position is $s(t) = t^3 - 6t^2 + 9t$. Find the velocity $v(t)$.

Wed Oct 27
At rest
TRY When is that particle at rest?

Thu Oct 28
Speeding up?
TRY Is the particle speeding up or slowing down at $t = 4$?

Fri Oct 29
Weekly Quiz
10 mixed problems: 8 on this week's skills + 2 review. Correct and log every miss.

WEEK 12 • Nov 1–Nov 5, 2027 — Unit 3: Applying Derivatives

Related Rates

Core: Write an equation linking the quantities, then differentiate with respect to time.

Mon Nov 1
Circles
TRY A circle's radius grows at 2 cm/s. How fast is its area growing when $r = 5$?

Tue Nov 2
Ladders
TRY A 10-ft ladder's foot slides away from a wall at 1 ft/s. How fast is the top sliding down when the foot is 6 ft out?

Wed Nov 3
Spheres
TRY A sphere's volume grows at 100 cm^3/s. How fast is the radius growing when $r = 5$?

Thu Nov 4
Squares
TRY A square's side grows at 3 cm/s. How fast is its area growing when the side is 10 cm?

Fri Nov 5
Weekly Quiz
10 mixed problems: 8 on this week's skills + 2 review. Correct and log every miss.

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WEEK 13 • Nov 8–Nov 12, 2027 — Unit 3: Applying Derivatives

Linearization & L'Hôpital's Rule

Core: Near a, $f(x) \approx f(a) + f'(a)(x-a)$.

Mon Nov 8

Linear approximation

TRY Use the tangent line at $x = 25$ to estimate $\sqrt{26}$.

Tue Nov 9

Over or under?

TRY Is that estimate of $\sqrt{26}$ too high or too low? Why?

Wed Nov 10

L'Hôpital: $\frac{0}{0}$

TRY Find $\lim_{x\to 0} \frac{e^x - 1}{x}$.

Thu Nov 11

HOLIDAY: Veterans Day

No class. Do this skill's 10 problems on the weekend:

L'Hôpital: $\frac{\infty}{\infty}$

TRY Find $\lim_{x\to\infty} \frac{\ln x}{x}$.

Fri Nov 12

Weekly Quiz

10 mixed problems: 8 on this week's skills + 2 review. Correct and log every miss.

WEEK 14 • Nov 15–Nov 19, 2027 — Unit 3: Applying Derivatives

The Mean Value Theorem & Extrema

Core: Extrema happen at critical points or endpoints.

Mon Nov 15

Mean Value Theorem

TRY For $f(x) = x^2$ on $[1, 3]$, find the c promised by the MVT.

Tue Nov 16

Critical points

TRY Find the critical points of $f(x) = x^3 - 3x$.

Wed Nov 17

Increasing & decreasing

TRY Where is $f(x) = x^3 - 3x$ decreasing?

Thu Nov 18

Absolute extrema

TRY Find the absolute max and min of $f(x) = x^3 - 3x$ on $[0, 2]$.

Fri Nov 19

Weekly Quiz

10 mixed problems: 8 on this week's skills + 2 review. Correct and log every miss.

Thanksgiving Break • Nov 22–Nov 26, 2027 • no class (not counted in the 34 weeks)

WEEK 15 • Nov 29–Dec 3, 2027 — Unit 3: Applying Derivatives

Concavity & the Second Derivative Test

Core: $f'' > 0$: concave up; $f'' < 0$: concave down.

Mon Nov 29

Concavity

TRY Where is $f(x) = x^3 - 6x^2$ concave up?

Tue Nov 30

Inflection points

TRY Find the inflection point of $f(x) = x^3 - 6x^2$.

Wed Dec 1

Second derivative test

TRY For $f(x) = x^3 - 3x$, is $x = 1$ a local max or a local min?

Thu Dec 2

Reading f'

TRY f' changes from positive to negative at $x = 4$. What happens to f there?

Fri Dec 3

Weekly Quiz

10 mixed problems: 8 on this week's skills + 2 review. Correct and log every miss.

WEEK 16 • Dec 6–Dec 10, 2027 — Unit 4: Optimization & Integration

Optimization & Curve Sketching

Core: Write the quantity to optimize in one variable, then find its critical points.

Mon Dec 6

Fencing

TRY 100 m of fence encloses a rectangle. What is the largest possible area?

Tue Dec 7

Open boxes

TRY Squares of side x are cut from the corners of a 12×12 sheet to make an open box. Find the x that gives the most volume, and that volume.

Wed Dec 8

Closest point

TRY Find the points on $y = x^2$ closest to $(0, 2)$.

Thu Dec 9

Curve sketching

TRY Find the local minimum of $f(x) = x^4 - 4x^3$.

Fri Dec 10

Weekly Quiz

10 mixed problems: 8 on this week's skills + 2 review. Correct and log every miss.

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WEEK 17 • Dec 13–Dec 17, 2027 — Unit 4: Optimization & Integration

Riemann Sums & Accumulation

Core: Add up (rate) × (small change) to find the total change.

Mon Dec 13
Left sums
TRY Estimate $\int_0^4 x^2\,dx$ with 4 equal left rectangles.

Tue Dec 14
Right sums
TRY Estimate the same integral with 4 equal right rectangles.

Wed Dec 15
Trapezoids
TRY Estimate the same integral with 4 equal trapezoids.

Thu Dec 16
Meaning
TRY $r(t)$ is a flow rate in gallons per minute. What does $\int_0^{10} r(t)\,dt$ mean?

Fri Dec 17
Weekly Quiz
10 mixed problems: 8 on this week's skills + 2 review. Correct and log every miss.

Winter Break • Dec 20–Dec 24, 2027 • no class (not counted in the 34 weeks)

Winter Break • Dec 27–Dec 31, 2027 • no class (not counted in the 34 weeks)

WEEK 18 • Jan 3–Jan 7, 2028 — Unit 4: Optimization & Integration

The Fundamental Theorem of Calculus

Core: $\int_a^b f(x)\,dx = F(b) - F(a)$, and $\frac{d}{dx}\int_a^x f(t)\,dt = f(x)$.

Mon Jan 3
Evaluating
TRY Find $\int_1^3 2x\,dx$.

Tue Jan 4
Polynomials
TRY Find $\int_0^2 (3x^2 - 1)\,dx$.

Wed Jan 5
FTC, part 1
TRY Find $\frac{d}{dx}\int_0^x \cos(t^2)\,dt$.

Thu Jan 6
FTC with the chain rule
TRY Find $\frac{d}{dx}\int_0^{x^2} \sin t\,dt$.

Fri Jan 7
Weekly Quiz
10 mixed problems: 8 on this week's skills + 2 review. Correct and log every miss.

WEEK 19 • Jan 10–Jan 14, 2028 — Unit 4: Optimization & Integration

Antiderivatives & Indefinite Integrals

Core: Reverse each derivative rule, and do not forget $+C$.

Mon Jan 10
Power rule
TRY Find $\int (4x^3 + 2)\,dx$.

Tue Jan 11
Trig
TRY Find $\int \cos x\,dx$ and $\int \sec^2 x\,dx$.

Wed Jan 12
Exponentials & $\frac{1}{x}$
TRY Find $\int \left(e^x + \frac{1}{x}\right)dx$.

Thu Jan 13
Initial conditions
TRY If $f'(x) = 6x$ and $f(1) = 5$, find $f(x)$.

Fri Jan 14
Weekly Quiz
10 mixed problems: 8 on this week's skills + 2 review. Correct and log every miss.

WEEK 20 • Jan 17–Jan 21, 2028 — Unit 4: Optimization & Integration

Mid-Year Review & Checkpoint

Core: Review Units 1–4 and practice working steadily through a full test.

Mon Jan 17
HOLIDAY: MLK Jr. Day
No class. Do this skill's 10 problems on the weekend:
Review: derivatives
TRY Find $\frac{d}{dx}\left(x \cdot e^{2x}\right)$.

Tue Jan 18
Review: related rates
TRY A circle's radius grows at 3 cm/s. How fast is the area growing when $r = 2$?

Wed Jan 19
Review: extrema
TRY Find the critical point of $x^2 - 6x$.

Thu Jan 20
Review: integrals
TRY Find $\int_0^{\pi} \sin x\,dx$.

Fri Jan 21
Mid-Year Checkpoint
Units 1–4 test. Then sort errors by topic in the error log.

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WEEK 21 • Jan 24–Jan 28, 2028 — Unit 5: Integration & Differential Equations

Integration by Substitution **Core:** Choose u as the "inside" function whose derivative also appears.

Mon Jan 24
Basic substitution
TRY Find $\int 2x(x^2+1)^4\,dx$.

Tue Jan 25
Linear inside
TRY Find $\int \cos 3x\,dx$.

Wed Jan 26
Exponentials
TRY Find $\int x \cdot e^{x^2}\,dx$.

Thu Jan 27
Definite integrals
TRY Find $\int_0^1 2x(x^2+1)^3\,dx$.

Fri Jan 28
Weekly Quiz
10 mixed problems: 8 on this week's skills + 2 review. Correct and log every miss.

WEEK 22 • Jan 31–Feb 4, 2028 — Unit 5: Integration & Differential Equations

Accumulation Functions & Integral Properties **Core:** $g(x) = \int_a^x f(t)\,dt$ accumulates area, and $g'(x) = f(x)$.

Mon Jan 31
Reversing limits
TRY If $\int_2^5 f(x)\,dx = 7$, find $\int_5^2 f(x)\,dx$.

Tue Feb 1
Adding intervals
TRY If $\int_0^3 f = 4$ and $\int_3^7 f = -1$, find $\int_0^7 f$.

Wed Feb 2
Accumulation functions
TRY If $g(x) = \int_1^x (t^2-4)\,dt$, find $g'(3)$.

Thu Feb 3
Odd functions
TRY Find $\int_{-2}^{2} x^3\,dx$.

Fri Feb 4
Weekly Quiz
10 mixed problems: 8 on this week's skills + 2 review. Correct and log every miss.

WEEK 23 • Feb 7–Feb 11, 2028 — Unit 5: Integration & Differential Equations

Slope Fields & Separable Equations **Core:** Separate the variables, integrate both sides, then use the initial condition.

Mon Feb 7
Checking solutions
TRY Is $y = e^{3x}$ a solution of $y' = 3y$?

Tue Feb 8
Slope fields
TRY For $\frac{dy}{dx} = x - y$, find the slope at $(2, 1)$.

Wed Feb 9
Separation of variables
TRY Solve $\frac{dy}{dx} = x \cdot y$ with $y(0) = 2$.

Thu Feb 10
Another separable equation
TRY Solve $\frac{dy}{dx} = \frac{2x}{y}$ with $y(0) = 3$.

Fri Feb 11
Weekly Quiz
10 mixed problems: 8 on this week's skills + 2 review. Correct and log every miss.

WEEK 24 • Feb 14–Feb 18, 2028 — Unit 5: Integration & Differential Equations

Exponential Models **Core:** If $\frac{dy}{dt} = ky$, then $y = y_0 e^{kt}$.

Mon Feb 14
Writing the model
TRY Solve $\frac{dP}{dt} = 0.05P$ with $P(0) = 200$.

Tue Feb 15
Doubling time
TRY With $k = 0.05$, how long does it take P to double?

Wed Feb 16
Decay
TRY A substance has a half-life of 10 years. Find k.

Thu Feb 17
Equilibrium
TRY For $\frac{dT}{dt} = -k(T-70)$ with $k > 0$, what temperature does T approach?

Fri Feb 18
Weekly Quiz
10 mixed problems: 8 on this week's skills + 2 review. Correct and log every miss.

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WEEK 25 • Feb 21–Feb 25, 2028 — Unit 5: Integration & Differential Equations

Average Value & Motion with Integrals

Core: Average value $= \frac{1}{b-a}\int_a^b f(x)\,dx$; $\int |v|\,dt$ is total distance.

Mon Feb 21

HOLIDAY: Presidents' Day

No class. Do this skill's 10 problems on the weekend:

Average value

TRY Find the average value of x^2 on $[0, 3]$.

Tue Feb 22

Displacement

TRY Find the displacement for $v(t) = t - 2$ on $[0, 4]$.

Wed Feb 23

Total distance

TRY Find the total distance for $v(t) = t - 2$ on $[0, 4]$.

Thu Feb 24

Position from velocity

TRY If $v(t) = 3t^2$ and $s(0) = 5$, find $s(2)$.

Fri Feb 25

Weekly Quiz

10 mixed problems: 8 on this week's skills + 2 review. Correct and log every miss.

WEEK 26 • Feb 28–Mar 3, 2028 — Unit 6: Applications of Integration

Area Between Curves

Core: Area $= \int_a^b (\text{top} - \text{bottom})\,dx$.

Mon Feb 28

Under a curve

TRY Find the area under $y = x^2$ from $x = 0$ to $x = 3$.

Tue Feb 29

Between curves

TRY Find the area between $y = x$ and $y = x^2$.

Wed Mar 1

Integrating in y

TRY Find the area between $x = y^2$ and $x = 4$.

Thu Mar 2

Finding intersections

TRY Find the area between $y = 2x$ and $y = x^2$.

Fri Mar 3

Weekly Quiz

10 mixed problems: 8 on this week's skills + 2 review. Correct and log every miss.

WEEK 27 • Mar 6–Mar 10, 2028 — Unit 6: Applications of Integration

Volumes with Cross Sections

Core: Volume $= \int_a^b A(x)\,dx$, where $A(x)$ is the area of a cross section.

Mon Mar 6

Square cross sections

TRY The base is under $y = \sqrt{x}$ on $[0, 4]$. Cross sections perpendicular to the x-axis are squares. Find the volume.

Tue Mar 7

Semicircles

TRY The base is between $y = x$ and the x-axis on $[0, 2]$. Cross sections are semicircles with diameter in the base. Find the volume.

Wed Mar 8

Triangular base

TRY The base is the triangle under $y = 2 - x$ in the first quadrant. Cross sections are squares. Find the volume.

Thu Mar 9

Setting up

TRY The base is under $y = e^x$ on $[0, 1]$ with square cross sections. Find the volume.

Fri Mar 10

Weekly Quiz

10 mixed problems: 8 on this week's skills + 2 review. Correct and log every miss.

WEEK 28 • Mar 13–Mar 17, 2028 — Unit 6: Applications of Integration

Volumes: Disc & Washer Methods

Core: Disc: $\pi \int R^2\,dx$. Washer: $\pi \int (R^2 - r^2)\,dx$.

Mon Mar 13

Discs

TRY Rotate the region under $y = \sqrt{x}$ on $[0, 4]$ about the x-axis. Find the volume.

Tue Mar 14

More discs

TRY Rotate the region under $y = x^2$ on $[0, 2]$ about the x-axis. Find the volume.

Wed Mar 15

Washers

TRY Rotate the region between $y = x$ and $y = x^2$ about the x-axis. Find the volume.

Thu Mar 16

About the y-axis

TRY Rotate the region between $y = x^2$ $(x \geq 0)$, the y-axis and $y = 4$ about the y-axis. Find the volume.

Fri Mar 17

Weekly Quiz

10 mixed problems: 8 on this week's skills + 2 review. Correct and log every miss.

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WEEK 29 • Mar 20–Mar 24, 2028 — Unit 6: Applications of Integration

Accumulation in Context

Core: Amount now = starting amount + $\int$ (rate in − rate out) dt.

Mon Mar 20
Net change
TRY A tank has 20 L. Water flows in at $r(t) = 10 + 2t$ L/min for 5 minutes. How much is in it then?

Tue Mar 21
In & out
TRY Water flows in at 8 L/min and out at 3 L/min for 6 minutes. How much does the amount change?

Wed Mar 22
When is it largest?
TRY Water flows in at $r(t) = 12 - 2t$ L/min and out at 5 L/min. When is the amount greatest?

Thu Mar 23
Meaning & units
TRY $r(t)$ is people per hour entering a park. What does $\int_0^2 r(t)\,dt$ mean?

Fri Mar 24
Weekly Quiz
10 mixed problems: 8 on this week's skills + 2 review. Correct and log every miss.

Spring Break • Mar 27–Mar 31, 2028 • no class (not counted in the 34 weeks)

WEEK 30 • Apr 3–Apr 7, 2028 — Unit 6: Applications of Integration

Free-Response Practice: Tables & Graphs

Core: Justify every answer with calculus language and the theorem you used.

Mon Apr 3
Derivatives from tables
TRY $v(2) = 10$ and $v(5) = 19$. Estimate $v'(3.5)$.

Tue Apr 4
Riemann sums from tables
TRY t: 0, 2, 5 and $r(t)$: 4, 6, 9. Find the left Riemann sum for $\int_0^5 r(t)\,dt$.

Wed Apr 5
Reading f'
TRY f' changes from negative to positive at $x = 1$. What happens to f there? Justify.

Thu Apr 6
MVT from tables
TRY v is differentiable, $v(2) = 10$ and $v(5) = 19$. Must $a(t) = 3$ at some time in $(2, 5)$?

Fri Apr 7
Weekly Quiz
10 mixed problems: 8 on this week's skills + 2 review. Correct and log every miss.

WEEK 31 • Apr 10–Apr 14, 2028 — Unit 6: Applications of Integration

Multiple-Choice Practice

Core: Know the core facts cold so you can save time for the hard problems.

Mon Apr 10
Key limit
TRY Find $\lim_{x\to 0} \frac{\sin x}{x}$.

Tue Apr 11
Derivative at a point
TRY Find the derivative of $(x^2 + 1)^3$ at $x = 1$.

Wed Apr 12
Key integral
TRY Find $\int_1^e \frac{1}{x}\,dx$.

Thu Apr 13
Sign changes
TRY On $(0, \pi)$, where does $f'(x) = \cos x - 0.5$ change sign?

Fri Apr 14
Weekly Quiz
10 mixed problems: 8 on this week's skills + 2 review. Correct and log every miss.

WEEK 32 • Apr 17–Apr 21, 2028 — AP Exam Review

AP Review: Limits & Derivatives

Core: Mix the skills from Units 1–3 and justify each answer.

Mon Apr 17
Review: limits
TRY Find $\lim_{x\to\infty} \frac{5x^3-2}{2x^3+x}$.

Tue Apr 18
Review: chain rule
TRY Find $\frac{d}{dx}\sqrt{1 + x^2}$.

Wed Apr 19
Review: implicit derivatives
TRY Find $\frac{dy}{dx}$ for $x \cdot y = 6$ at $(2, 3)$.

Thu Apr 20
Timed set: 10 in 20 min
TRY Find the tangent line to $y = \ln x$ at $x = 1$.

Fri Apr 21
Weekly Quiz
10 mixed AP-style problems, timed (25 min).

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WEEK 33 • Apr 24–Apr 28, 2028 — AP Exam Review

AP Review & Practice Exam

Core: Practice a full exam, then learn from every error.

Mon Apr 24

Review: integrals

TRY Find $\int_0^1 (x^3 + x)\,dx$.

Tue Apr 25

Review: differential equations

TRY Solve $\frac{dy}{dx} = y$ with $y(0) = 4$.

Wed Apr 26

Review: volume

TRY Rotate the region under $y = x$ on $[0, 3]$ about the x-axis. Find the volume.

Thu Apr 27

Practice Exam

Full-length AP Calculus AB practice exam, timed.

Fri Apr 28

Error-log review

Redo every missed problem from the exam; write the key idea for each.

WEEK 34 • May 1–May 5, 2028 — AP Exam Review

Final Checkpoint & Celebration

Core: Show what you have learned and go into the AP exam with confidence.

Mon May 1

Light review

TRY Find the derivative of x^4 at $x = 2$.

Tue May 2

Mixed warm-up set

TRY Find $\int_0^2 3x^2\,dx$.

Wed May 3

Final Checkpoint

Covers the whole year.

Thu May 4

Checkpoint review

Go over solutions together; students explain one problem each.

Fri May 5

Celebration & reflection

Certificates, progress review, and goals for Calculus BC or college math.

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ANSWER KEY

Answers to the Daily "Try" Problems

Week 1
Mon: 4
Tue: -1 and 1
Wed: No: the one-sided limits differ
Thu: ∞

Week 2
Mon: 10
Tue: 6
Wed: $\frac{1}{4}$
Thu: 3

Week 3
Mon: Yes
Tue: $k = \frac{3}{2}$
Wed: Removable
Thu: Yes, by the IVT

Week 4
Mon: 3
Tue: 0
Wed: 0
Thu: 0

Week 5
Mon: 2.1
Tue: $2x$
Wed: $y = 6x - 9$
Thu: No: the graph has a corner

Week 6
Mon: $12x^2 - \frac{1}{\sqrt{x}}$
Tue: $5x^4 - 1$
Wed: $\frac{1}{(x+1)^2}$
Thu: 11

Week 7
Mon: $\cos x - \sin x$
Tue: $\sec^2 x$
Wed: $e^x(x + 1)$
Thu: $\frac{1}{4}$

Week 8
Mon: $15(3x + 1)^4$
Tue: $2x\cos(x^2)$
Wed: $-2e^{-2x}$
Thu: 12

Week 9
Mon: $-\frac{x}{y}$
Tue: $-\frac{3}{4}$
Wed: $\frac{1}{1+x^2}$
Thu: $\frac{1}{4}$

Week 10
Mon: 0
Tue: $3x^2 \sin x + x^3 \cos x$
Wed: $\frac{2x}{x^2+1}$
Thu: $y = x + 1$

Week 11
Mon: At $t = 5$, the water is decreasing at 3 L per minute
Tue: $3t^2 - 12t + 9$
Wed: $t = 1$ and $t = 3$
Thu: Speeding up: $v(4) > 0$ and $a(4) > 0$

Week 12
Mon: 20π cm^2/s
Tue: $\frac{3}{4}$ ft/s
Wed: $\frac{1}{\pi}$ cm/s
Thu: 60 cm^2/s

Week 13
Mon: 5.1
Tue: Too high: $\sqrt{x}$ is concave down
Wed: 1
Thu: 0

Week 14
Mon: $c = 2$
Tue: $x = \pm 1$
Wed: $-1 < x < 1$
Thu: Max 2, min -2

Week 15
Mon: $x > 2$
Tue: At $x = 2$
Wed: Local min ($f''(1) = 6 > 0$)
Thu: A local maximum

Week 16
Mon: 625 m^2
Tue: $x = 2$; volume 128
Wed: $\left(\pm\sqrt{\frac{3}{2}}, \frac{3}{2}\right)$
Thu: At $x = 3$; $f(3) = -27$

Week 17
Mon: 14
Tue: 30
Wed: 22
Thu: Total gallons in the first 10 minutes

Week 18
Mon: 8
Tue: 6
Wed: $\cos(x^2)$
Thu: $2x\sin(x^2)$

Week 19
Mon: $x^4 + 2x + C$
Tue: $\sin x + C$ and $\tan x + C$
Wed: $e^x + \ln|x| + C$
Thu: $f(x) = 3x^2 + 2$

Week 20
Mon: $e^{2x}(1 + 2x)$
Tue: 12π cm^2/s
Wed: $x = 3$
Thu: 2

Week 21
Mon: $\frac{(x^2+1)^5}{5} + C$
Tue: $\frac{\sin 3x}{3} + C$
Wed: $\frac{e^{x^2}}{2} + C$
Thu: $\frac{15}{4}$

Week 22
Mon: -7
Tue: 3
Wed: 5
Thu: 0

Week 23
Mon: Yes
Tue: 1
Wed: $y = 2e^{x^2/2}$
Thu: $y = \sqrt{2x^2 + 9}$

Week 24
Mon: $P = 200e^{0.05t}$
Tue: $\frac{\ln 2}{0.05} \approx 13.86$
Wed: $k = -\frac{\ln 2}{10} \approx -0.0693$
Thu: 70

Week 25
Mon: 3
Tue: 0
Wed: 4
Thu: 13

Week 26
Mon: 9
Tue: $\frac{1}{6}$
Wed: $\frac{32}{3}$
Thu: $\frac{4}{3}$

Week 27
Mon: 8
Tue: $\frac{\pi}{3}$
Wed: $\frac{8}{3}$
Thu: $\frac{e^2-1}{2}$

Week 28
Mon: 8π
Tue: $\frac{32\pi}{5}$
Wed: $\frac{2\pi}{15}$
Thu: 8π

Week 29
Mon: 95 L
Tue: It rises 30 L
Wed: $t = 3.5$ min
Thu: The number of people who enter in the first 2 hours

Week 30
Mon: 3
Tue: 26
Wed: Local min: f' changes from $-$ to $+$
Thu: Yes, by the Mean Value Theorem

Week 31
Mon: 1
Tue: 24
Wed: 1
Thu: $x = \frac{\pi}{3}$

Week 32
Mon: $\frac{5}{2}$
Tue: $\frac{x}{\sqrt{1+x^2}}$
Wed: $-\frac{3}{2}$
Thu: $y = x - 1$

Week 33
Mon: $\frac{3}{4}$
Tue: $y = 4e^x$
Wed: 9π

Week 34
Mon: 32
Tue: 8

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